Particles as Localized Topological Defects of Native Overlap Transport: Finite Theorems, a Native Decision Test and Mass Obstructions

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Abstract

We state and test the matter hypothesis of Observer Patch Holography (OPH): stable massive matter is a localized topological defect of native overlap transport. A bounded core is surrounded by records that approach a reference regime, while the transport around the core carries a loop holonomy that admitted local repairs cannot erase. The rest mass is the excess rest energy of the defect over the reference, produced by its excess maintenance dynamics and read in the same clock and energy dictionary as its propagation. Finite theorems supply the seed and the target. A fixed nonzero cycle holonomy forces a strictly positive residual mismatch, and under declared premises the holonomy takes values in a compact gauge structure of type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ whose one-generation matter module carries the Standard Model charges. Neither result localizes, moves or weighs a defect. A direct test on the native two-load repair process rejects its simplest candidate. A double load survives n seam attempts with probability $(5/6)^n$, every classical configuration invariant is constant, the quantum collision subspace decays at the same rate, and the tested repair costs are not monotone energies. Exact obstructions exclude trace-one record averages, affine static family features and the tested record-memory laws as mass prescriptions. A calibrated action recovers supplied masses as a readout control. No native defect sector, maintenance-energy law or measured spectrum is derived. A separate supplied Weyl/contact action now has one moving, spectrally protected bound pair with complete charge, energy and spectral readouts and a native two-level clock attachment. Its band is topologically trivial, its coordinate rest/kinetic mass readings disagree under the same speed, and its interaction prevents exact removal of individual Weyl pockets. The missing physical joins are stated as constructions with stopping tests.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation

nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3 + 1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/BASIS.md

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join

needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

Finite C/E campaign. A versioned finite family tests both complete C and E instruments, their tagged classical mixtures, positive-probability identity holds and bounded classical memory with a positive uniform seam-exploration floor. On the existing 78-state carrier, the continuing observer retains a finite record buffer and explicitly exports displaced records or halts when full. Exact six-attempt executions preserve complete instrument probabilities and untouched-reference marginals. The transition kernel specifies every finite record history; the exponentially growing history tree is not materialized. Growing exports remain a resource, and preparation, randomization, access, storage and delivery services remain supplied.

The adaptive observer also has a constructed stationary reference. Its source/current-sign quotient has a unique recurrent class of 129 of 132 states. Lifting that exact law through the actual complete record instrument gives 128 positive record blocks for C and 192 for E, with 4,218 positive source/record diagonal entries in either case and exact stationary residual zero. This is the finite source/buffer marginal at repair probability one, not the growing history; faithfulness requires its stated support restriction. The delayed member needs its own reference construction.

Two exact obstructions sharpen the next joins. Each law's complete active branch $*$-algebra is the full matrix algebra on both the 78-state source and its 66-state binary shell. Thus a source-only Hermitian charge commuting with every branch is scalar; this does not exclude forward-invariant subspaces, general expectation-conserved observables, history-dependent charges or controlled lifetimes. Separately, the frozen real-operation grammar cannot implement a nontrivial modular action of the constructed nontracial reference on its fixed full support algebra: a specified input and dimensionless modular parameter give normalized trace-distance error at least $1/2$. The separating observable is a mathematical witness, not an admitted native detector. This is not an exclusion of every clock or encoding.

These results extend the component foundations and leave the integrated joins open. They sharpen four of them: a versioned extension and refinement class beyond the finite contract; a realization of the observer's declared services; accessible noncommuting algebras with implemented clock operations; and an explicitly specified defect sector or controlled lifetime outside the excluded commuting-charge class. The campaign adds no independent physical comparison and neither freezes nor excludes the full executable hypothesis.

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md

Same-action finite evidence. A supplied three-dimensional Weyl/contact CAR action now has one isolated two-fermion bound band whose wavepacket moves, carries the conserved CAR number charge and has complete contact/scattering spectral records. Its positive contact section makes the band topologically trivial. An exact rational curvature enclosure disagrees with the rest/kinetic mass relation using the same speed parameter; these are coordinate energies and masses, not identified particle species. One invariant pair and dark-reference subspace is attached by a supplied isometry to a native two-level corner. Its record-derived modular clock and complete herald, continuation and energy records are executed with a finite unconditional error bound. This does not encode the whole spatial field.

On the same local action, signed CAR regional restrictions, exact local energy conservation and finite quantum Gibbs/entropy identities now have explicit witnesses. They retain all local number sectors, references and detector outcomes. Regional bare-energy modular identification fails, and the contact interaction leaks from any proper Weyl momentum pocket into its complement. Finite thermal positivity and KMS identities do not remove that coupling or establish a chiral continuum. These results reduce specific finite obligations in the existing premise map; physical geometry, Lorentz stress, common normalization and integrated acceptance remain open. The current four construction targets are native defect protection, common energy/kinetics, the interacting chiral field and finite-theorem integration.

This paper's hypothesis and place in the tower

This paper is the matter half of layer 5. The gauge half proposes that gauge structure is the transport of overlap data and classifies the compact Lie type of that transport under complete-response premises [6]. The present hypothesis is that stable massive matter is a localized topological defect of the same native overlap transport, whose rest energy is its excess maintenance dynamics over the reference regime, read in the common clock and energy dictionary (Hypothesis 1.1).

What is proved. The finite results are unconditional statements about stated finite objects: the frustration theorem at fixed holonomy [3], the topology of the native carrier complex, the exact decision test on the native two-load repair process, the collision projector identity, the record-cost theorem for the corrected code, and the normalization, affine-family, rate and scale obstructions for the tested mass prescriptions. The one-generation matter decoder, the Higgs channels and the interaction dictionaries are conditional implications of their declared spin lift, exterior module and actions.

Open joins. The executable model must supply a native defect sector with a protecting invariant under its full admitted law, the excess maintenance energy with kinetics and stress, defect transport with creation, annihilation, fusion and decay rules, an interacting chiral $3+1$ description containing massless radiation, and a frozen dictionary compared with the matched spectrum (Section 8).

Refutation. A proof that every invariant of the full admitted law on the declared carrier and refinement class is constant, including its quantum and complete-history sectors, removes the native defect mechanism for that law. A protected sector whose excess maintenance quantity is not one energy with the common signal speed of its propagation rejects the maintenance reading of mass. A frozen defect dictionary that fails the matched spectrum rejects that law and dictionary.

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1 The defect hypothesis

OPH models observations through finite accessible algebras, retained records and local repair. The executable observer construction supplies explicit carriers, source operators, instruments and a native repair law [7, 8], and the observer layer supplies the record and measurement theorems on which a detector reading rests [2]. A particle in this setting is a statement about a continuing observable response: preparations, currents and detector records must exhibit one excitation, with its quantum numbers, propagation and interactions, in a specified state and clock regime. A finite internal basis vector, memory level, channel decay rate or agreement record does not have that meaning by itself, and no fundamental particle object behind the observers' statistics is assumed.

1.1 Statement

Hypothesis 1.1 (Localized defects of native overlap transport)**.** *Stable massive matter is a localized defect of the native overlap and transport structure, in the following sense.*

(i) Core and reference. *There is a reference regime of the native process and a bounded core outside which the records approach that regime.*

(ii) Protecting invariant. *The transport around the core carries a loop holonomy, a nontrivial invariant relative to the reference, that admitted local repairs cannot erase. The obstruction is to trivializing the transport relative to the reference; observers agree on every measured event.*

(iii) Transport. *Moving the core transports the invariant.*

(iv) One law. *Creation, annihilation, fusion and decay follow from the same native law. Freezing selected seams or supplying punctures to protect an assigned label changes the law and does not count.*

(v) Rest energy. *The rest mass is the excess rest energy of the defect over the reference, produced by its excess maintenance dynamics and read in the same clock and energy dictionary as its propagation and interactions. In a justified local relativistic regime this gives* $E_{\mathrm{rest}} = mc^2$ *with one common signal speed* c*, the speed of the massless response of the same construction. No conversion factor is fitted per species.*

(vi) Unstable species. *Unstable particles are metastable or internal excitations of defects, with decay channels and widths derived from the same law.*

(vii) Radiation. *Massless radiation and the gauge response arise from the same construction, as transport or collective excitations of the overlap structure.*

(viii) Identity. *Species, charges, spin and statistics, and families are properties of defect classes and of their transport.*

The hypothesis is the matter clause of the executable-model hypothesis stated above. It specifies a mechanism and its observable consequences. It is not implied by mass–energy equivalence: a real free scalar field has a contractible field target and trivial vacuum topology, yet its one-particle states have relativistic massive dispersion [9]. A dispersion relation alone therefore does not reveal a topological sector. Whether an effective particle is microscopically a defect is a question about the source realization.

The hypothesis names parameters explicitly. Native attempt counts, host runtime, solver counters and modular parameters are distinct from calibrated observer clocks, and repair counts, entropy production and resource use are distinct from energy. The maintenance-to-energy map in clause (v) must be derived, including kinetic normalization and any reference or regulator subtraction. Different schedules or implementations can change resource counts without changing the claimed physical response; an energy must be invariant under such changes.

1.2 Observable criteria

Each clause has a test that a finite or refined execution can pass or fail. The tests are conjunctive: they must hold for one excitation of one continuing source process, with one state, one clock and one set of accessible operations.

Part of the hypothesis	Concrete test
Defect identity	Exhibit an actual native state or observable sector and a boundary charge invariant under the admitted repairs; derive its coefficient group and transport rules.
Persistence and localization	Retain the full law and show that the core persists or has a controlled lifetime, with reference-like response outside it; distinguish active recovery from autonomous protection.
Maintenance and energy	Derive the excess action or energy of the defect relative to the reference using a calibrated local clock, with all kinetic terms from that same process.
Mass and propagation	In a justified local relativistic regime, recover the same rest energy from propagation and dispersion; the rest-mass relation uses the common signal speed, not a conversion fitted per species.
Particle identity	Obtain spin and statistics, charges, current couplings and any family structure on those same defects or excitations, before comparing mass ratios.

A candidate identification names one excitation or sector of one continuing source process before its mass comparison: the reference regime, local preparation, support, defect invariant, admitted repairs, propagation and allowed creation and annihilation channels. The same object must account for currents, detector response, kinetic normalization and its mass observable. Representation labels, memory levels, decay rates and benchmark field poles cannot be pooled into a spectrum by choosing the closest numerical row.

2 What the finite theorems supply

2.1 Frustration at fixed holonomy

The mathematical seed of the hypothesis is the cycle-obstruction theorem of the consensus layer [3]. Let A be an abelian group, $G = (V, E)$ a finite connected graph with oriented edges, $b_e \in A$ an edge offset and $w_e > 0$ an edge weight. The affine overlap equations $x_v - x_u = b_e$ have a global solution exactly when every cycle holonomy $\sum_{e \in C} \varepsilon_C(e) b_e$ vanishes.

Theorem 2.1 (Persistent frustration at fixed holonomy [3])**.** *Define the mismatch cost*

$$\Phi_b(x) = \sum_{e:u \to v} w_e \, \mathbf{1}[x_v - x_u \neq b_e], \qquad x : V \to A.$$

If some cycle holonomy of b is nonzero, then $\min_x \Phi_b(x) > 0$.

Proof. The cost takes values among the finite subset sums of the positive weights, so its minimum is attained. A zero minimum would solve every edge equation, contradicting the cycle criterion. □

Repairs of the vertex assignments x cannot remove this frustration while b is fixed, and it persists under changes of b inside its cohomology class. The theorem lacks four things that the hypothesis needs. It has no update rule for b; a physical update that changes the compatibility data outside its class can remove the obstruction. It does not localize the frustration in a bounded core. It does not move it. Its residual $\min_x \Phi_b$ is a cost of the whole compatibility problem, not an excitation energy above a reference ground state of its own sector.

On a two-dimensional complex the holonomy and its localization are related by an elementary identity. Let K be a finite oriented two-complex, b an A-valued one-cochain and $F = \delta b$ its face curvature, $F(f) = b(\partial f)$.

Lemma 2.2 (Loop holonomy is enclosed curvature)**.** *For every two-chain c, the holonomy of b around ∂c equals $F(c)$. If $H_1(K;A) = 0$, every closed edge loop is such a boundary, so every loop holonomy is the curvature it encloses. If K is a closed oriented surface with coherently oriented faces, then $\sum_f F(f) = 0$. If $b = \delta n$ is an exact difference, then $F = 0$ and every loop holonomy vanishes.*

Proof. The first statement is the definition of the coboundary. A one-cycle with vanishing homology class bounds a two-chain. The sum of all coherently oriented faces of a closed surface has zero boundary. Finally, $\delta\delta n = 0$. □

On a closed surface with vanishing first homology, a nonzero holonomy around a bounded core therefore requires nonzero curvature inside the core, and the total curvature vanishes: cores come with compensating curvature elsewhere. For a nonabelian structure group the holonomy of a based loop is defined up to conjugation, and its conjugacy class is the invariant that a defect charge would carry.

2.2 A quantitative frustration instance

Frustration produces a strictly positive gap on the executable carrier. Let D_σ be the sign-twisted seam derivative on the connected observer-visible finite local domain, and let $H_\sigma = D_\sigma^* D_\sigma$ use the declared unit-counting measure.

Proposition 2.3 (Exact finite source-Hamiltonian gap)**.** *The signed-incidence formula makes H_σ integer symmetric and positive semidefinite. A kernel vector would supply a sign-consistent coloring on each connected component. The declared reversing seam transport frustrates every triangle, so the connected complex has no such coloring and the signed incidence rank is* 8,662*. Therefore* $\ker H_\sigma = 0$*. With maximum degree* 12*, the determinant bound gives*

$$\lambda_{\min}(H_\sigma) \geq 24^{-8661} > 0.$$

A sparse numerical refinement gives $\lambda_{\min} = 0.11753670336118684$ *with relative residual* 1.49×10^{-14}.

The determinant floor follows because the positive integer determinant is at least one and every eigenvalue is at most 24. The decimal is measured, with a residual check and no interval enclosure. This is the $\mathbb{Z}_2$ analogue of Theorem 2.1 with frustration on every face. It is a global gap of a declared operator. It does not select a localized defect, a physical field fiber or a mass.

2.3 The classification target

The gauge layer supplies the structure group in which defect holonomies would take values [6]. Under its complete-response and endogenous-transport premises, a faithful compact commutator-closed twelve-port response on which proper carrier rechartings act by inner automorphisms has the single compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Under the stated faithful-representation and central-kernel hypotheses the global group is $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$. A declared spin lift and exterior module give the one-generation matter module with exact Standard Model charges, anomaly cancellation, the common central kernel $\mathbb{Z}_6$ and the three charged-fermion Higgs channels (Section 5). The carrier premises and the closure hypothesis are stated with that classification [6]; the results below use them only through it.

These are the classification target for defect charges. If the hypothesis holds, the charge of a stable defect is a holonomy class in this group and the matter labels of the decoder are properties

of defect classes and their transport. How the representation labels of the module arise from a defect sector is part of the particle-identity join. The finite gauge identification fixes allowed representations; it does not fix their excitation energies.

2.4 Native topology and the finite flux criterion

The native carrier is the oriented boundary complex of an icosahedron: twelve ports, thirty seams and twenty faces, with Euler characteristic two [4]. The full two-dimensional complex has vanishing H_1, while H_2 does not vanish. The seam graph alone has cycle rank $30 - 12 + 1 = 19$; its cycles all bound in the two-complex. A nontrivial bundle needs actual transition data, and by Lemma 2.2 a nonzero loop holonomy needs face curvature.

Native port loads supply none. The load contrast $b_{ab} = n_a - n_b$ on a seam is an exact difference, so every face circulation is zero; the exact receipt checks all twenty faces on every state of the two-load domain of Section 3. This excludes flux in the actual load-gradient map only. It does not exclude other bundles, enlarged states or quantum sectors.

Declared $\mathbb{Z}_6$ link variables with a puncture pair give a separate flux experiment. Six flux classes, realized by two-puncture witnesses, are distinguished by the spectra of the twisted seam-adjacency operators up to the charge-conjugation pairing $k \leftrightarrow 6 - k$; regauging by a coboundary leaves each spectrum fixed, flux classes add under chain addition, and one edge-midpoint refinement to 42 ports, 120 seams and 80 faces reproduces the six classes. The links and punctures are supplied state and update premises. Unrestricted local link changes can erase the pair. The flux criterion therefore shows what a protected sector would look like on this carrier without exhibiting one in the native process.

3 The native decision test

The simplest local candidate in the native process is a double occupation of one port in the unchanged total-load-two C/E repair process on the twelve ports. Its domain has 78 configurations: twelve double loads and 66 binary states with two unit loads. The test retains all 720 raw outcome maps and compares the same state transitions, transport and cost observables. The invariant test uses the support of all native seams. Lifetime and activity numbers use an explicitly supplied independent uniform-seam and fair-coin probe schedule, which is not the production feedback policy. No physical duration is assigned to an attempt.

Test under the unchanged native law	Exact finite result	Consequence
Configuration sectors	Twelve transient double-load classes feed one closed irreducible binary class of 66 states	A scalar configuration label preserved by every admitted branch is constant on the 78-state domain
Local double-load persistence	Under independent uniform choice of the 30 seams, survival is $(5/6)^n$ and the mean lifetime is six seam attempts	The candidate splits into two unit loads; it is not a protected defect transported by this law
Quantum collision sector	Both complete C/E instruments satisfy $\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$, with no binary-to-collision feed	The same survival law holds for arbitrary quantum inputs and untouched references
Continuing binary transport	Nonzero current probability 5/33 per attempt in the uniform binary stationary law; every binary event has zero drop of $\Phi = \frac{1}{2}\sum_p n_p^2$	Repair activity and motion are not potential decrease
Proposed edge cost	The binary edge-disagreement cost takes the values 8 and 10, with native transitions in both directions	It is not a monotone maintenance-energy law of this process
Changed-law control	Removing the five seams at one port gives binary closed classes of sizes 11 and 55	A frozen port creates sectors by changing the allowed law

Several entries have short explanations. The native seam rule sets the two endpoint loads to the floor and ceiling of their mean, with a fair coin for an odd remainder. A double load next to an empty port therefore splits whenever one of its five incident seams is chosen, which happens with probability 5/30 per attempt at each of the twelve locations; it never hops as a unit. Two unit loads on the icosahedron disagree across ten seams when they are not adjacent and across eight when they are, and native moves change adjacency in both directions. At a binary seam the rule either waits or swaps, so Φ never changes while loads keep moving. Holding one port leaves either that port loaded with the other load free on eleven ports, or that port empty with $\binom{11}{2} = 55$ placements.

Proposition 3.1 (Collision decay for quantum inputs)**.** *Let $\{K_h\}$ be the complete uniform C or E instrument on the 78-state domain, $K_h = M_h/\sqrt{60}$ over the thirty seams, two coins and retained raw outcomes, including zero-probability outcomes. Then*

$$\sum_h K_h^\dagger P_{\mathrm{coll}} K_h = \frac{5}{6} P_{\mathrm{coll}}$$

for the projector P_{coll} onto the twelve double-load configurations. For every state ρ on the domain and an untouched reference,

$$\mathrm{Tr}\big[(P_{\mathrm{coll}} \otimes I)(\Phi^n \otimes \mathrm{id})(\rho)\big] = \Big(\frac{5}{6}\Big)^n \mathrm{Tr}\big[(P_{\mathrm{coll}} \otimes I)\rho\big],$$

where Φ is the unread one-attempt channel. No persistent quantum subspace is supported wholly on the collisions under this unread evolution.

Proof. The Heisenberg identity is verified exactly on all retained branches; its effect matrix is diagonal with entries 50/60 on collisions, and no branch feeds a binary state into a collision. The dual channel is linear, so $\Phi^{*n}(P_{\mathrm{coll}}) = (5/6)^n P_{\mathrm{coll}}$, and tensoring with the identity on the reference preserves the identity. A persistent collision-supported subspace would keep its collision weight equal to one. □

These results reject the proposed autonomous configuration defect and the tested automatic cost-to-energy identifications: attempt counts, the quadratic load potential and the edge-disagreement cost. They reject quantum persistence wholly within the collision subspace. They do not exclude other quantum, complete-history or federated defects, another repair law, a common energy offset, or the general hypothesis. A replacement candidate must identify its actual source sector and allowed dynamics before any mass comparison.

3.1 The actively corrected code

The actively corrected matching code is a different persistent object. It uses six fixed nonedge matching words; coherent recovery preserves its logical information, and actual port rotations act on it. Without recovery, one uniform-seam C or E attempt takes the normalized code projector out of the code with probability 31/108 or 55/108, respectively. This is a one-step escape from that preparation, not an all-state lifetime. With the declared recorded recovery, the logical state and its references are preserved and the complete record law has the product form

$$\rho \longmapsto \rho \otimes p(\text{records}).$$

Proposition 3.2 (Record costs cannot split logical states)**.** *For a channel of this product form and any observable c of the records alone, the expectation* $\mathrm{Tr}[(I \otimes c)(\rho \otimes p)] = \mathrm{Tr}(\rho)\,\mathbb{E}_p[c]$ *is independent of the logical state ρ.*

Proof. The trace factorizes over the tensor product. □

The exact receipt checks all 720 raw branches. The one-port cost, each native edge cost and the quadratic load potential compress on the code to $I_6/6$, $I_6/3$ and I_6. A fixed cost of seams or records alone therefore cannot give different masses to different logical states. A common energy offset is not excluded, and no physical offset is derived. Persistence here is active recovery selected by existing records, not autonomous protection, and the code carries no exhibited topological charge. Its fixed matching table and the separately derived equivariant code table must not be identified for a nonscalar observable or a new interaction.

4 Mass from maintenance

4.1 Requirements

A rest mass can be the excess rest energy of a localized excitation over the reference state in a regime where energy, clock and spatial dispersion are defined. To identify it with repair maintenance, the source must derive how the excitation changes an intrinsic action or energy readout, or the same clocked response; subtract the reference contribution; and account for cutoff, volume and renormalization dependence. Propagation and interactions are then tested with that same identification. Attenuation can govern lifetime or decoherence instead of inertial mass, and the response calculation must decide which.

Counting alone cannot supply the scale.

Proposition 4.1 (Count-rate no-go)**.** *An oriented register with twenty-four slots, a twenty-four-stage automaton, or a decomposition into twenty-four formal update steps determines no physical frequency, Hamiltonian gap or mass scale.*

Proof. Rescaling every physical generator by $H \mapsto \lambda H$, $\lambda > 0$, preserves the register, transition graph, grading and settled quotient while multiplying all generator gaps and rates by λ. □

The number twenty-four plays no role in the argument, and a repair iteration counter is likewise not an operational clock. A physical rate requires a source-derived clock instrument, a calibrated comparison line and a residual bound. Once such a line exists, a scale follows exactly.

Theorem 4.2 (Conditional determinant-clock attachment)**.** *Let L_{24} be a one-dimensional source-derived operational clock line with a distinguished calibrated positive gap vector ℓ_{24} of norm Δ_{24}. Suppose there is a refinement-natural, quotient-visible norm-preserving line isomorphism $\Theta_e : L_{24}^{\otimes 3} \to \det M_e$ to the physical charged-lepton determinant line, with the element compatibility $\Theta_e(\ell_{24}^{\otimes 3}) = \det(M_e)$. Then $g_e := |\det M_e|^{1/3} = \Delta_{24}$. If the element-compatible map rescales norms by a separately derived factor $\kappa > 0$, then $g_e = \kappa^{1/3}\Delta_{24}$.*

Proof. The distinguished tensor-cube vector has norm Δ_{24}^3. Element compatibility and norm preservation give $|\det M_e| = \Delta_{24}^3$. □

Writing a norm-preserving line isomorphism without the distinguished-vector equality constructs neither the clock line nor the attachment. Data that fix only centered logarithms leave the scale free.

Theorem 4.3 (Absolute-scale underdetermination)**.** *Any readout factoring only through a centered log triple (ℓ_1, ℓ_2, ℓ_3), $\sum_i \ell_i = 0$, and its difference invariants leaves the common scale $g > 0$ of $m_i = g\, e^{\ell_i}$ free.*

Proof. For every $c \in \mathbb{R}$, $e^c \operatorname{diag}(e^{\ell_i})$ has the same centered logarithms and differences, determinant e^{3c}, and masses scaled by e^c. □

A rest energy appears to a comoving observer as a clock rate. On a declared Lorentz module and mass shell, free plane waves persist under the declared translation flow, and their phase along their own frame has proper-time derivative m. For $m > 0$ the internal-clock periods are $(2\pi/m)\mathbb{Z}$; the zero-mass clock is constant and has no fundamental period. This is conditional kinematics, without a source-selected mass or clock, and it states the target that a maintenance rate must meet: the same m must appear in the internal clock, in the dispersion and in the kinetic normalization.

Kinetic and mass terms must come from the same source. For a four-component Dirac convention with positive kinetic metrics,

$$Y_{s,\mathrm{can}} = Z_L^{-1/2} Y_s Z_R^{-1/2} / \sqrt{Z_H};$$

a literal all-left Weyl bilinear has a transpose on the left normalization factor. A record matrix can change a coordinate without changing the mass: $Z_L = Z_R = R > 0$ and $Y_s = g_s R$ give $Y_{s,\mathrm{can}} = g_s I/\sqrt{Z_H}$ and $\det(pR - hg_sR) = \det(R)(p - hg_s)^3$. A hierarchical Gram matrix by itself need not produce a hierarchical physical vertex. Eliminating an invertible heavy block also changes the Higgs derivative,

$$\partial_H(A - BD^{-1}C) = A_H - B_H D^{-1}C - BD^{-1}C_H + BD^{-1}D_H D^{-1}C.$$

Repair contraction and coherent propagation are different dynamics. Passive stationary fixed-reference schedules have an exact rank-two Schur response obstruction, including eight zero crossing weights among the 22 seams, and positive whitening preserves the null direction. The uncorrected recorded 66-state channel has no invariant fifteen-state work code in the tested class. Its ten-mode response conditioned on an incident event has operator norm $\lambda_C = 7/10$ or $\lambda_E = 2/5$.

Proposition 4.4 (Contraction has no direct unitary intertwiner)**.** *Let R have operator norm $\lambda < 1$, let U be unitary and let J be any linear map. Then $\|UJ - JR\|_{\mathrm{HS}} \geq (1-\lambda)\|J\|_{\mathrm{HS}}$. In particular, the conditioned native response has no nonzero exact direct intertwiner to the unmeasured closed quadratic Weyl evolution.*

Proof. $\|UJ\|_{\mathrm{HS}} = \|J\|_{\mathrm{HS}}$ and $\|JR\|_{\mathrm{HS}} \leq \lambda\|J\|_{\mathrm{HS}}$; apply the triangle inequality. □

The proposition does not exclude a field coupled to detectors, an environment or interactions; a complete dephasing-channel control realizes the same uniform contraction. A contraction modulus describes attenuation in its declared event parameter, and the argument of a transfer eigenvalue describes a phase only with an unwrapping and clock convention. Neither becomes a rest mass by taking a logarithm.

4.2 Obstructions for the tested mass prescriptions

The following results are theorems about specified classes of uncalibrated prescriptions. Each excludes its class as the charged mass law. None excludes other repair laws on the carrier.

The complete record memory. A complete twelve-state family memory uses the native events with probabilities $p = 1/12$ and $q = 1/6$, a specified reset state and declared half-turn updates. All 128 native raw labels are combined with all twelve old memory states; conditioning on successful events is not substituted for this 1,536-branch law. The stationary projector average

$$Y = \sum_k \pi_k P_k \succeq 0, \qquad \operatorname{tr} Y = 1,$$

has coefficient ratios $1 : 195.11703757 : 3579.82624571$. A two-angle record inverse cancels the finite-pulse bias exactly; the resulting numerical allowance is conditional on its floating-point and mixing estimates. Gating the same memory with the complete thirteen-outcome matter instrument

$$M_0 = \cos(t\sqrt{C}), \qquad M_a = iF_a \frac{\sin(t\sqrt{C})}{\sqrt{C}}, \quad a = 1, \ldots, 12,$$

with $C = \sum_a F_a^\dagger F_a$ built from the matter lift of the trace-orthonormal current frame, and with both components of each Higgs tensor, gives sector-dependent coefficients without empirical mass inputs. At the declared $t = \pi/5$ the ratios for the channels matched after construction are $1 : 271.446 : 6071.036$ (up), $1 : 311.762 : 6822.880$ (down) and $1 : 675.782 : 30775.333$ (charged lepton). The entire declared catalog $t \in \{\pi/5, \pi/3, \pi/2\}$ fails the charged hierarchy comparison; its zero coefficient at $t = \pi/2$ is retained. This finite candidate uses supplied control services and a declared architecture. It is a tested prescription, not a derivation of the spectrum.

Proposition 4.5 (Normalized-feedback scale bound)**.** *For a probability average of rank-one projectors on a three-dimensional family space, $Y \succeq 0$, $\operatorname{tr} Y = 1$ and $1/3 \leq \lambda_{\max}(Y) \leq 1$. With common canonical kinetics, unit-modulus neutral Higgs tensors and a common Higgs norm v, the three family action masses in each sector sum to v. Any two sector maxima have ratio at most three.*

Proof. The nonnegative eigenvalues sum to one, so their largest lies between their mean and their sum. The specified Higgs tensors multiply each family singular value by v. Apply the bounds to both sectors. □

This is a sector-normalization obstruction under the common-coupling and canonical-kinetic hypotheses. It does not bound a smallest eigenvalue away from zero and does not exclude within-sector hierarchies. A sector-dependent strength or a different kinetic response changes the hypotheses and needs its own derivation. Positive trace-one matrices can differ in eigenbasis, so the bound implies nothing about mixing or CP violation. Unconditional event-duty pulses change the normalization and must use the complete correlated record law: for the reset projector P_2 the C-duty response is $Z_C = Y - (1-p)P_2$, not pY. The always, C, E and CE policies have traces $1, p, q, pq$; their sector-maximum spans are approximately $1.0233, 2.0572, 1.1528, 2.3695$, and the E and CE responses have rank two. These policies are excluded as the charged mass prescription.

Static family features. The native quadratic comparison uses $Y_u = aF$, $Y_d = bF$, $Y_e = JF$ with

$$a = \log(25/8), \quad b = \log(8/5), \quad J = a - b, \quad v = a + b.$$

The family matrix F has eigenvalues

$$\frac{61}{450}, \qquad \frac{41}{225} \pm \frac{7\sqrt{5}}{750},$$

and the direct positive prescription has ratios approximately $1 : 1.1903 : 1.4982$, far from a charged hierarchy. Complex affine coefficients do not repair the class.

Proposition 4.6 (Affine quadratic-family exclusion)**.** *Let $Y = uI + wF$, $u, w \in \mathbb{C}$, with singular values $m_{\min} \le m_{\mathrm{mid}} \le m_{\max}$. With $\varphi = (1+\sqrt{5})/2$,*

$$m_{\max} \le \varphi^2 m_{\mathrm{mid}} + \varphi m_{\min}.$$

Proof. For ordered eigenvalues $f_1 < f_2 < f_3$ of F, the complex points $z_i = u + wf_i$ are collinear, and $f_3 - f_1 = \varphi^2(f_2 - f_1)$ gives $z_3 - \varphi^2 z_2 + \varphi z_1 = 0$. Solving this relation for each point and applying the triangle inequality bounds any modulus by φ^2 times one of the others plus φ times the third. The middle point is a convex combination of the endpoints. Applying these bounds to the largest modulus gives the sorted inequality. □

Conditional on $m_{\mathrm{mid}}/m_{\min} \le 207$, the bound gives $m_{\max}/m_{\min} \le 543.551$, whereas the broad charged-lepton pole-ratio windows $[206, 207]$ and $[3470, 3485]$ require at least 3470. The adopted $\overline{\mathrm{MS}}$ comparison at M_Z has lepton ratios about 210.7 and 3578 [13], and the inequality excludes that comparison as well. The bound is not a universal heavy-to-light bound: a light eigenvalue near zero can make that single ratio arbitrarily large. The class has a fixed quadratic action, so its anticommutator poles are state independent; an occupation-versus-gap factor of two, a common clock rescaling or a basis change cannot change their ratios.

Other explicit laws. A five-memory feedback law gives ratios near $1 : 30.84 : 452.62$; a two-rotation memory gives $1 : 382.23 : 5909.43$; the admitted auxiliary-dressing class bounds its two smallest subthreshold lepton poles by a ratio below 32. A near numerical ratio found by searching many prescriptions is not a blind prediction; the selection ancestry and all admitted alternatives must be retained.

Symmetry and entropy selectors. Residual symmetry does not select the ratios. Realize the quadrupole carrier W_5 as traceless symmetric real 3×3 matrices. Invariance under a threefold or fivefold rotation about n forces $A = \alpha(nn^T - I/3)$, with a double eigenvalue. The twofold fixed locus is three-dimensional and admits simple spectra, and every subgroup of A_5 other than the trivial group, C_2 and V_4 contains a threefold or fivefold rotation. No residual stabilizer pins an isolated simple-spectrum orbit. For six antipodal pair weights $p_i = (1 + ra_i)/6$ at fixed index of coincidence, every global minimizer of the full uniform-reference relative entropy $D(p\|\mathbf{1}/6)$ for $0 < r \leq \sqrt{60}$ is a permutation of one high and five equal low weights, the standard maximum-entropy branch at fixed index of coincidence [12]. Its quadrupole spectrum is $(-2/\sqrt{15}, -2/\sqrt{15}, 4/\sqrt{15})$ up to scale, with the fivefold-axis stabilizer. No amplitude rule therefore yields three distinct masses through an attachment that reads only these eigenvalues. The leading cubic truncation selects the same degenerate orbit. The strict full-support minima of the quartic truncation have sorted-gap ratio φ or a double eigenvalue, and the quartic truncation reaches the central charged-lepton gap ratio only at a target-attached amplitude on its closed boundary, where eight of twelve port probabilities vanish; that is not a prediction. These exclusions concern the stated reference, constraint and readout.

4.3 Calibrated control and target-informed constructions

A calibration packet supplies nine running charged Yukawas, mixing parameters and the Higgs scale in stated schemes. The finite action checks its internal mass and current identities, and separate source-record experiments and a moving-field readout recover the supplied running parameters with a fixed probe design and a blind inverse. This is a readout test: the Hamiltonian retains the empirical inputs, and decoder blindness does not remove that ancestry. A running action parameter is not automatically a laboratory pole mass.

Several numerical constructions have target ancestry and are retained only as such: fixed closure-coordinate running charts for quarks, the electroweak bosons and Higgs criticality; an eight-register charged-lepton response architecture; a down-type Clebsch assignment ranked against measurements; and the equal-weight block realization of the Koide balance. They are conditional constructions: their discrete choices and some attachments have target ancestry. Freezing them does not erase that ancestry. Their transmutation recipe uses supersymmetric one-loop coefficients $(33/5, 1, -3)$, whereas the stated matter content of three fifteen-state generations and one Higgs doublet gives $(41/10, -19/6, -7)$ in the same normalization, and their energy unit is the Planck energy formed with measured G. A square of renormalization and Higgs maps that returns the same declared tuple along both routes is a declared-map packaging identity, not a construction of source-derived comparison maps. The current packet supplies no OPH result about Higgs naturalness. None of these constructions is a derivation, and the measured particle spectrum is not derived.

5 Matter and interaction readout from the finite response

The finite response determines more than the abstract gauge Lie type. Its matrix generators support an exact, conditional matter decoder and local interaction tensors. A representation is distinct from a physical particle: a particle identification also consumes its spin, statistics, propagator, interaction and observer-clock interpretation.

5.1 From response operators to a fifteen-state matter module

Let a response generator on the six-dimensional carrier have blocks $D = \mathrm{diag}(A, B)$, with $A \in \mathfrak{u}(3)$ and $B \in \mathfrak{so}(3)$. Write $B = \widehat{w}$. The declared spin lift and central trace redistribution give

$$L(D) = \mathrm{diag}\left(A, -\frac{i}{2} w \cdot \sigma - \frac{\mathrm{tr}\, A}{2} I_2\right).$$

This is a Lie-algebra homomorphism. The complete twelve-direction response and all sixty-six bracket pairs are checked exactly. The central direction, after clearing denominators and choosing the weak-positive orientation, has primitive integer character

$$q = \mathrm{diag}(-2, -2, -2, 3, 3).$$

The lift and trace redistribution are hypotheses. A native weak rotation through 2π is the identity on its vector carrier and acts as minus the identity on the lifted doublet. Thus the lifted module is not a subrepresentation of the original six-dimensional response. Global descent and a physical current interpretation require the corresponding additional identifications.

Proposition 5.1 (Conditional matter decoder)**.** *For the lifted five-dimensional one-body action, the representation*

$$M = \Lambda^2 \mathbb{C}^5 \oplus \Lambda^4 \mathbb{C}^5$$

has dimension fifteen. Its charge and quadratic Casimir operators identify the following irreducible components without a field-name input table.

Component	Color	Weak dimension	$Y = q/6$	Dimension
Q	3	2	1/6	6
u^c	$\overline{3}$	1	$-2/3$	3
d^c	$\overline{3}$	1	1/3	3
L	1	2	$-1/2$	2
e^c	1	1	1	1

Proof. Split $\mathbb{C}^5 = V_c \oplus V_w$, of dimensions three and two and charges -2 and 3. Then

$$\Lambda^2(V_c \oplus V_w) = \Lambda^2 V_c \oplus (V_c \otimes V_w) \oplus \Lambda^2 V_w,$$

with integer charges $-4, 1, 6$. The degree-four summands are $\Lambda^3 V_c \otimes V_w$ and $\Lambda^2 V_c \otimes \Lambda^2 V_w$, of charges $-3, 2$. Color and weak Casimirs separate the representations, and Cartan weights distinguish the conjugate color character. Exact projectors give the same answer after a rational change of basis mixing unlike charge sectors. □

The auxiliary exterior construction has two additional invariant lines, the vacuum and top exterior degree. Removing them is part of the declared fifteen-state module. Charge traces alone do not remove neutral invariant states. Its even auxiliary parity also does not establish physical fermion statistics. Physical two-component Weyl fields require an independent CAR algebra and spin interpretation. The anomaly-free charge table and common central kernel are conditional representation results; they do not identify laboratory species or select a generation count.

5.2 One Higgs doublet and the exact interaction tensors

For two same-chirality physical Weyl fields, the antisymmetric spinor contraction makes the internal coefficient symmetric. The Lorentz-scalar internal pair space is consequently $\mathrm{Sym}^2 M$, of dimension 120. A color-Casimir kernel and weak-Casimir projector identify one doublet of hypercharge $-1/2$ and two of hypercharge $+1/2$. Their constituent integer-charge pairs are $(-4,1)$, $(-3,6)$, and $(1,2)$. These are interpolating channels; their existence supplies no bound-state pole, condensate or elementary-boson commutation relation.

Proposition 5.2 (Conditional one-doublet interaction)**.** *Declare one scalar doublet H of hypercharge $1/2$. Each of the three channels above has a one-dimensional invariant contraction space. They give*

$$Qu^cH, \qquad Qd^cH^*, \qquad Le^cH^*.$$

The same scalar doublet and its conjugate occur in all three expressions.

Proof. Restrict the exact pair generators to each color-singlet doublet. The tensor sum of that doublet generator and the generator of H, or its conjugate, has a one-dimensional common kernel. Equivalently, its coefficient matrices C_α satisfy, for each of the twelve source currents,

$$T^tC_\alpha + C_\alpha T + \sum_\beta (T_H)_{\beta\alpha} C_\beta = 0.$$

Here the conjugate action is $-T_H^t$ in the skew-Hermitian convention. The physical pairing kernel $\epsilon \otimes C_\alpha$ is antisymmetric. The identity $S^t\epsilon S = \epsilon$ for $S \in \mathrm{SL}(2,\mathbb{C})$ establishes its Lorentz scalar character. The executable checks the rotation and boost generators as well as all current identities. □

For three declared families the coefficients are three complex 3×3 matrices, with 27 complex entries before flavor-basis redefinitions. The pairing convention is

$$\mathcal{L}_{\mathrm{pair}} = \frac{1}{2}\epsilon_{AB}\psi_A^t K(H)\psi_B + \mathrm{h.c.}$$

up to an overall sign. With canonical kinetic normalization and $H = (0, v)$, its singular values are tree-level mass parameters; $v = v_{\mathrm{SM}}/\sqrt{2}$ in the conventional electroweak normalization. Exact matrix-unit tests cover arbitrary complex family coefficients. Their values, the Higgs potential and its vacuum are not selected by these identities. The optional invariant $(LH)(LH)/\Lambda$ has six complex symmetric family coefficients and gives $M_\nu = \kappa v^2/\Lambda$. It is an additional dimension-five effective interaction, with no selected coefficient, scale or ultraviolet completion.

If both family factors are identified with the same unbroken irreducible rank-three source representation and the Higgs is family-neutral, the exact commutant has dimension one. In that branch each Yukawa intertwiner is yI_3, giving equal singular values. A nonscalar background can split the band; choosing that background is an additional physical hypothesis. The source family-band theorem therefore does not imply the observed hierarchy.

5.3 Scope of the readout

A declared coherent filter on the native six-dimensional response admits an exact invertible twelve-control map into the same compact algebra; the spin/exterior and Higgs tensor maps above apply after this change of controls. It must not be identified with state cooling. For a rank-three projector P, the cooling channel $\Phi(\rho) = P\rho P + \mathrm{tr}((I-P)\rho)P/3$ has dual $\Phi^*(B) = PBP + \mathrm{tr}(PB)(I-P)/3$,

which is not the Hamiltonian map $h \mapsto PhP$. Its classical recorded branches lose the relative phase of the two triplets.

The executable calculations reconstruct the response, representation projectors and interaction tensors from repository-local source operators, with exact rational or algebraic identities separated from numerical residuals. They use no measured particle masses. They supply neither a physical chiral field net nor a source-selected Higgs vacuum, Yukawa matrix or mass pole. A relaxation gap, finite cavity frequency or commuting internal modular offset requires its own mass-shell identification before it can be called a physical rest mass.

6 Further supporting results

6.1 Currents and carrier covariance

The native protected geometric object has type $3 \oplus 3'$. Twelve ports count interfaces, not independent Lie generators. The tested native exchange response closes at seventeen currents. A specified law instead closes at $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$, of dimension twelve, and a coherent Hamiltonian filter has an exact invertible control map into the same algebra. This is a conditional control construction; ordinary C/E state repair is not shown to select the physical gauge response. The full exchange-plus-class-sum controller has eighteen directions; a fixed-bias long-time limit has six observable directions, and a joint bias and update limit with bounded drive approaches twelve.

The carrier recharting premise acts by inner automorphisms on the current algebra, and the real part of the colour generators implements rotations about the port axes. These facts establish carrier covariance. They do not establish which transformations observers recover as physical spatial rotations: equivalent finite-group representations on different factors need not be one physical action, and declaring one action to be both frame rotation and colour transport does not establish their independence. A rank-three local carrier frame is not a proof of three-dimensional physical propagation on a federation.

6.2 Finite interaction identities

With a declared scalar quartic, breaking scale and gauge couplings, the finite electroweak-breaking theorem identifies the vacuum shell, its one-dimensional stabilizer, three gauge-orbit Hessian zero directions and the tree gauge-mass ratio. The slot dictionaries then give photon charge eQ, neutral shape $g_Z(T_3 - Q\sin^2\theta_W)$, and charged shifts $g_W/\sqrt{2}$ on doublets and zero on singlets. These are identities of the declared infinitesimal action and pairing, with supplied constants; they construct no physical current operators, mixing, decay amplitudes or Fermi limit.

For a declared complex algebra with $[X_i, X_j] = iq\epsilon_{ijk}B_k$, the Pauli identity reads

$$(\sigma \cdot X)^2 = (X \cdot X)I - q\,\sigma \cdot B,$$

which in the stated orbital and spin normalization is $g = 2$. It constructs no gauge potential, electron, anomalous moment or physical coupling.

The ten-row exterior census follows from the 32 subsets of five modes: bidegrees (c, w) have multiplicity $\binom{3}{c}\binom{2}{w}$ and charge $-2c+3w$, and removing vacuum and top degree leaves ten bidegrees. A supplied anomaly-free chiral mask and central weights recover the same six-element kernel. The finite family interface selects rank three under its single-object, faithful-action and operational-cost hypotheses; a separate conditional argument from intrinsic CP capability and weak-sector ultraviolet completeness gives $3 \le N_g \le 5$. Neither attaches a physical family field or excludes extra light sectors.

On the declared carrier $C \oplus W = \mathbb{C}^3 \oplus \mathbb{C}^2$, with $H = h_0 e_3 + h_1 e_4$, the top-form contractions $B_u = [Q \wedge H \wedge u^c]_5$ and $B_d = [(\iota_{\overline{H}} Q) \wedge d^c]_5$ give

$$U(H) = I_C \otimes \begin{pmatrix} h_1 \\ -h_0 \end{pmatrix}, \qquad D(H) = I_C \otimes \begin{pmatrix} -\overline{h}_0 \\ -\overline{h}_1 \end{pmatrix}, \qquad U^\dagger U = D^\dagger D = \|H\|^2 I_C, \quad U^\dagger D = 0.$$

The full doublet has unit-magnitude color Clebsches, and the channel coefficients g_u, g_d are not fixed by normalizing these tensors. On a family attachment $F \otimes M$ that restricts every supplied operator to $I_F \otimes A$, the result is $Y_u = g_u I_F$ and $Y_d = g_d I_F$, preserved by products, adjoints, inverses, positive functional calculus and family-independent heavy-mode elimination. This bounded direct-contraction model produces no hierarchy.

6.3 Flavor constraints

Species are identified by represented charges, Casimirs and interaction channels. Sorting a nondegenerate singular spectrum names mass eigenstates; it does not derive three families or their weak couplings. For canonical Dirac matrices with $U_{sL}^\dagger Y_s U_{sR} = \mathrm{diag}(y_s)$, the charged-current matrix is $V = U_{uL}^\dagger U_{dL}$, and a physical CP test uses a rephasing-invariant quantity such as the Jarlskog invariant [54, 15]. Yukawa matrices simultaneously real in one weak basis have zero CP-odd invariant, and a common matrix up to sector scalars has trivial charged-current mixing for a simple spectrum. Orientation reversal of a cycle phase is physical time reversal only with its own antiunitary intertwiner, and a classical cycle affinity is a log probability ratio, not a coherent U(1) phase.

For a positive Hermitian circulant square-root response $C = aI + bR + \overline{b}R^2$, $R^3 = I$, $b = are^{i\delta}$, with eigenvalues the nonnegative square roots of the masses,

$$Q := \frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{1}{3} + \frac{2}{3} r^2,$$

since $\mathrm{tr}\, C = 3a$ and $\mathrm{tr}\, C^2 = 3a^2 + 6|b|^2$. Thus $Q = 2/3$ exactly when $r = 1/\sqrt{2}$ [10, 11]. The identity selects no positive branch, phase, absolute scale or physical response; the balance condition that realizes $Q = 2/3$ is a premise with measured-triple ancestry.

Conditional on the registered source signature containing no typed datum that separates the stated centered-spread orbit, a generic quark package does not define a unique quark mass spectrum. Granting an ordered profile shape, write its centered logarithms as $\sigma_u \widehat{\ell}_u$ and $\sigma_d \widehat{\ell}_d$. Independent positive changes of (σ_u, σ_d) preserve the certified gauge representations, anomaly traces and invariant Higgs lines, so those certificates leave at least an $(\mathbb{R}_{>0})^2$ fiber in this granted subfamily. Without the shape hypothesis, sector means and shapes are also undetermined. Representative independence does not select the datum: a constant map on a bridge fiber descends to its quotient whatever its value, and a target-derived spread keeps that ancestry after descent. Two finite exclusions complete the picture. The candidate reciprocal-ray law requires a product of adjacent log-gap ratios equal to one, whereas common-scale $\overline{\mathrm{MS}}$ Yukawas at M_Z give approximately 0.83532 [13]. Directly identifying two real A_5 residual axes with the Cabibbo angle fails: the least nonzero angle is about 20.905°, with sine 0.356822, against the measured approximately 0.225. Under additional alphabet, measure-balance and register-faithfulness hypotheses a finite enumeration selects the Clebsch set $\{1/3, 1, 3\}$. Ranking its six generation assignments against measurements is target-informed. The selected assignment $y_b = y_\tau$, $y_s = y_\mu/3$, $y_d = 3y_e$ is the Georgi–Jarlskog texture [46], and it fails the adopted joint lattice comparison [14].

6.4 Response readout tools

The response diagnostics test a field interpretation once a defect sector exists. The finite interacting CAR response determines its inverse and frequency derivative from all Lehmann residues, with canonical sum-rule and positivity admission. Kinetic and mass forms transform together under a change of interpolators, so raw matrix eigenvalues are not invariant mass data. A charge-resolved census of the supplied 180-mode Weyl–Nambu symbol retains signed and dark modes and unresolved frequency lists.

For one charged L/e^c pair, a finite neutral-reference experiment reconstructs the complex two-mode transfer from complete detector and clock records. Charge-compatible source pulses prepare a fixed-total-charge-zero reference, and all measured continuation branches are kept. On a supplied principal logarithm window and canonical quadratic response class, the inverse recovers the generator, residues and joint z-axis momentum and neutral-Higgs multiplier derivatives. The retained 304 contexts have 4,864 terminal outcomes. Withheld settings and a second duration give generator residuals at most about 6.40×10^{-8} per unit of reconstructed clock coordinate and complete-output distances about 4.36×10^{-8}; both durations repeat one elementary source word, with composition defect about 2.23×10^{-16}. These are numerical diagnostics. Phase-sensitive controls detect opposite coupling signs hidden by Rabi marginals, and withheld momenta detect an otherwise indistinguishable two-hop kinetic control. The experiment reconstructs a component of a supplied action; it selects no mass coefficient and introduces no particle candidate.

A finite abelian quantum link has an electric–matter interaction and an exact operator-Gauss sector on a restricted four-state code. A Euclidean overlap calculation on the fifteen-state representation checks the modified chiral projector and gauge covariance on a nonconstant nonabelian background, with a stated small-plaquette locality domain; exact gauge-invariant constructions of this type are known for abelian chiral lattice theories [21, 22, 23]. The free spatial implementation has conditional Weyl–Higgs dispersion on a supplied background; its cubic staggered continuum contains two Dirac tastes and both chiralities [100, 101], and the naive lattice has eight balanced Weyl nodes. Deleting unwanted nodes is not a decoupling proof.

7 One interacting bound pair and its remaining physical joins

The finite construction now follows one excitation through protection, motion, charge, energy and a complete spectral readout. It uses the supplied three-dimensional CAR lattice action

$$H = T + UP, \qquad T = d\Gamma(h), \qquad h(k) = -c\sum_{j=1}^{3}\sigma_j \sin k_j, \qquad P = \sum_x n_{x0}n_{x1}, \qquad c = \tfrac{1}{8}, \quad U = 1,$$

on the complete antisymmetric two-fermion sector. Here P is the orthogonal projector onto on-site singlets. The lattice, CAR dictionary, couplings and Hamiltonian model-time convention are inputs. Periodic boxes resolve momentum; the open boxes used for regional observations have the same local terms but different boundaries. Infinite volume below keeps lattice spacing fixed. Repulsively bound lattice pairs are established physics [97]; the contribution is their explicit same-action attachment to the source-observation construction.

Proposition 7.1 (Isolated pair band on the declared lattice)**.** *Let $b = 2\sqrt{3}c$ and $U > b$. On a periodic box with V sites, exactly V eigenvalues of H lie in the upper band, all at least U, while every remaining eigenvalue is at most b. Thus the separation is at least $U - b$. There is one upper-band state in each total-momentum fiber.*

Proof. The symbol obeys $h(k)^2 = c^2 \sum_j \sin^2(k_j) I$, so $\|T\| \le b$. A single hop separates an on-site pair, hence $PTP = 0$. The quadratic form of H on $\operatorname{Ran} P$ is exactly U, while on its orthogonal complement it is that of T and is at most b. Min–max gives the two bounds and the band count. In each momentum fiber P has rank one. □

Writing $s = (0, 1, -1, 0)/\sqrt{2}$ and $d_K(k) = s/\sqrt{V}$, the free fiber is $B_K(k) = h(k) \otimes I + I \otimes h(K-k)$. Its upper pole and contact residue satisfy

$$g(E, K) = \frac{1}{V} \sum_k \langle s, (E - B_K(k))^{-1} s \rangle, \qquad Ug(E, K) = 1,$$

$$Z(K) = \left[\frac{U^2}{V} \sum_k \langle s, (E - B_K(k))^{-2} s \rangle \right]^{-1}, \qquad \psi_K = U \sqrt{Z(K)} (E - T_K)^{-1} d_K.$$

The resolvent is positive above the free spectrum and g is strictly decreasing there. The isolated pole is therefore unique. The receipt keeps the complementary spectral weight $1 - Z$, including all multiplicities and all physical states; it does not condition on contact detection.

The same resolvent gives relative localization. If $\Pi_{\ge R}$ selects relative Manhattan distance at least R, a hop changes that distance by at most one, and the first R terms of the Neumann series cannot contribute. For integers $R \ge 0$,

$$\|\Pi_{\ge R} \psi_K\|^2 \le \min \left\{ 1, \left(\frac{U}{U - b} \right)^2 \left(\frac{b}{U} \right)^{2R} \right\}.$$

This is spectral protection and localization under the closed declared Hamiltonian, not protection against arbitrary native C/E branches. The positive contact overlap fixes a global phase for the band. Continuously reducing c to zero preserves the separating gap and contracts the band to on-site pairs. Its line bundle is topologically trivial; no nontrivial native boundary holonomy is assigned by this construction.

The retained $L = 4$ calculation Fourier-includes all 64 band states into the complete 8,128-dimensional position-space CAR sector. An independently assembled hopping/contact Hamiltonian verifies that inclusion, and its evolution agrees with band propagation. A predeclared wavepacket changes the complete site-contact/separated outcome law by about 0.08986 at model time 16. Every relative and spin component remains present. On-site pair creation $b_x^* = a_{x0}^* a_{x1}^*$ obeys

$$[N, b_x^*] = 2b_x^*, \quad (b_x^*)^2 = 0, \quad [b_x, b_x^*] = 1 - n_{x0} - n_{x1}, \quad [b_x^*, b_y^*] = 0 \quad (x \ne y).$$

These are charge-two, even-parity, hard-core composite operators in the supplied CAR theory. The conserved number current is checked independently on the full finite action. Electromagnetic charge, Standard Model representations, generations and Lorentz spin remain unassigned. The Weyl hopping does not conserve standalone internal spin; the exact singlet statement below concerns its specified momentum.

7.1 One energy difference, one clock corner, and two mass readings

At $K_\eta = (\pi, \pi, \pi)$, $h(K_\eta - k) = h(k)$, and the total-spin operator kills s. Thus $Hd = Ud$ and $Z(K_\eta) = 1$. On each even periodic box of side at least four, $r(k) = \sqrt{2} \sin(k_x) d(k)$ is normalized, orthogonal to d and satisfies the physical exchange constraint. It has $Pr = Tr = 0$. The invariant space $S = \operatorname{span}\{r, d\}$ therefore has $H|_S = \operatorname{diag}(0, U)$. Both levels have number two and the same total momentum; r is not a vacuum.

The source clock maps native occupations $\{1,2\}$ and $\{7,2\}$ to (r,d) by an explicit supplied isometry. Their product-reference weights are 200/7505 and 64/7505, so the heralded corner has success mass 264/7505 and normalized density

$$\rho_S = \mathrm{diag}(25/33, 8/33) = e^{-\beta H|_S}/\operatorname{Tr}(e^{-\beta H|_S}), \qquad \beta U = \log(25/8).$$

Complete source executions retain both herald outcomes, native C/E continuation, leakage, the record-derived clock and energy readout. Their unconditional half-trace error bound is below 0.018762; it is never divided by the herald probability. The Heisenberg model coordinate is $t = -\beta s$ for modular parameter s. Faithfulness is only on M_2; neither the full spatial isometry nor a native full-field Gibbs preparation has been executed, and physical seconds and energy units remain open.

For the same band in infinite volume, let κ be the curvature $\partial_{K_j}^2 E$ at K_η. Resolving the singlet/triplet blocks in second-order perturbation theory gives

$$\kappa = \frac{2c^2}{U}\left\langle \cos^2 k_j \left[1 + \frac{4c^2(\sum_l \sin^2 k_l - \sin^2 k_j)}{U^2 - 4c^2 \sum_l \sin^2 k_l}\right]\right\rangle_{\mathrm{BZ}}.$$

Indeed, at K_η the first derivative couples d to the triplet through $c\cos k_j (I \otimes \sigma_j)$; the direct second-derivative expectation vanishes. Inverting the triplet operator yields the displayed positive denominator. This is a Brillouin-zone integral at fixed spacing, not a continuum limit.

Put $a = c^2/U^2$ and $m_n = \binom{2n}{n}/4^n$. The positive geometric series for that denominator gives the exact rational lower sum

$$R_N = 1 + 16a \sum_{n=0}^{N} (4a)^n \sum_{i+j+k=n} \frac{n!}{i!j!k!}(m_i - m_{i+1}) m_{j+1} m_k, \qquad R = \frac{U\kappa}{c^2},$$

with $0 \le R - R_N \le 4a(12a)^{N+1}/(1-12a)$. The bound follows from $\sum \sin^2 k_l \le 3$ and $\langle \cos^2 k_x (\sin^2 k_y + \sin^2 k_z)\rangle = 1/2$. At the declared parameters already $17/16 \le R \le 14/13$; the order-16 enclosure gives $R \simeq 1.06904778$ and kinetic coordinate mass $1/\kappa \simeq 59.86636$, whereas $U/c^2 = 64$. Under one clock/rod conversion $t_{\mathrm{phys}} = \tau t$, $x_{\mathrm{phys}} = \ell x$, the ratio $E\kappa/c^2$ is invariant. Such calibration cannot repair this particular same-speed relativistic mismatch. It does not exclude other scalings, reference states or candidate actions.

The energy split is also evaluated on ψ_K itself: $E_{\mathrm{int}} = UZ(K)$, $E_{\mathrm{kin}} = \langle \psi_K, T_K \psi_K\rangle$, and $\partial E/\partial U = Z(K)$ at fixed c, K. This is the response to the supplied action coupling, not a derivation of U from native repair maintenance or apparatus work.

7.2 The interacting taste boundary

Let Q retain a proper, exchange-symmetric momentum pocket in the $K = 0$ fiber, with $w = \langle d, Qd\rangle \in (0,1)$. The free action commutes with Q, while the contact projector gives exactly

$$\|[H, Q]\| = U\sqrt{w(1-w)}.$$

To see this, decompose $d = Qd + (1-Q)d$ into its two orthogonal normalized directions; the commutator has a two-dimensional off-diagonal block of that magnitude. On the $L = 6$ box the eight sign-of-cosine pockets have $w = 1/8$ each, contain the eight Weyl nodes, and are invariant under $k \mapsto -k$. A state $Qd/\sqrt{w}$ leaks into all seven omitted pockets under the same interaction. The squared short-time leakage coefficient is 7/64, and the complete omitted-pocket probability is about 0.26706 at model time two. Removing U removes this leakage. Dropping pockets is therefore

not an exact interacting chiral restriction of this action. This excludes neither another regulator nor a controlled decoupling limit.

The same complete physical fiber admits a faithful finite Gibbs density, positive Euclidean reflection Gram matrices and the finite KMS identity for its declared even observables. These matrix facts do not establish a spatial-continuum reconstruction, physical vacuum or chiral gauge theory. The executable sources, complete records and proof boundaries are retained in the public `bound_pair`, `particle_clock`, `field_regions`, `field_stress` and `field_tastes` model packages.

8 The open joins

The joins below are conjunctive. Each must hold on the same continuing source process, and each has a test that stops its branch when it fails.

Native defect sector. Exhibit a state or observable sector of the unchanged law, including its quantum, complete-history and federated or refined extensions, with a boundary invariant preserved by every admitted branch; derive its coefficient group, show localization with reference-like response outside a bounded core, and distinguish autonomous protection from active recovery. A proof that every admitted-branch invariant on the declared carrier and refinement class is constant, as on the 78-state domain, removes the native defect mechanism for that law. A sector that exists only after freezing seams or supplying links and punctures belongs to a changed law and is tested as a new candidate.

Maintenance energy, kinetics and stress. Derive the excess energy of the defect over the reference from the same process, with a calibrated local clock, its kinetic terms and its stress, so that the same energy enters the modular and stress dictionary used by the gravitational layer [5]. A maintenance quantity that changes under admissible rescheduling while the observable response is unchanged is not an energy. A rest energy from maintenance that differs from the rest energy read from dispersion with the common signal speed rejects the maintenance reading of mass for that law.

Transport and identity. Show that moving the core transports its invariant, and derive the rules for pair creation, annihilation, fusion and decay from the same law. Obtain spin and statistics, for example from defect exchange, together with charges, currents and any family structure on the same defects. A charge that moves only through supplied link updates, or labels that are attached without a transport derivation, fail this join.

Interacting chiral description. Establish locality, a state and vacuum, the chiral measure and anomaly control, dynamical gauge interactions, and controlled volume and continuum limits, with massless radiation and gauge response arising as transport or collective excitations of the same construction. Extra regulator species, dark modes and complementary modes stay in the census. A proof that the admitted native regulator class keeps balanced chiralities in every refinement limit excludes the chiral spectrum for that class.

Frozen dictionary and matched spectrum. Freeze the source, observation map, calibration and uncertainties before the comparison, retaining a hash, date, decision rule, all tested alternatives and prior target exposure. The charged-spectrum benchmark is nine running charged Yukawas and the four independent CKM parameters in compatible conventions; neutrino masses and their

mechanism are a further obligation. Compare ratios within a stated scheme and scale, then sector strengths, mixing and the operational absolute scale. A source-only GeV value is equivalent to a source-only dimensionless clock ratio, because the conversion through the defining cesium frequency is exact in SI units. A failed frozen comparison rejects that law and dictionary; changing the rule creates a new recorded candidate, and registration after exposure to known masses makes a postdiction.

Fixed boundaries. There is no strictly positive global vacuum mass gap in the intended infinite-volume theory with a massless photon, so sector gaps and massive thresholds name their state, sector and limit; a positive repair relaxation gap or finite-cavity spacing is not a physical mass gap. Quark masses require renormalization, matching and confinement observables, and confined gluons are not observable free particles. Unstable masses require resonance conventions, a pole on a stated sheet or a controlled line shape [18]. Charged sectors with massless gauge fields require an infrared treatment instead of an isolated one-particle pole [19, 20]. Under the hypothesis, conjugate defect classes supply antiparticles; a matter asymmetry additionally requires a physical CP-odd current, its state, generator, clock and thermal history.

Reproduction

The source operators, invariant projectors, Higgs tensors, native decision experiment, corrected-code control, record-memory protocol and hierarchy diagnosis are executable in the public repository with the observer construction [7, 1]. Their receipts distinguish exact algebra, floating-point diagnostics and empirical comparisons. A hash check verifies custody; it does not replace execution or extend a theorem's physical scope.

A Finite flavor constraints from a settlement rule

The constraints below concern the icosahedral carrier and a strict-descent settlement rule. That rule differs from the native seam rule of the executable observer, which sets the endpoint loads to the floor and ceiling of their mean and assigns an odd remainder by a fair coin; at load difference one the native rule waits or swaps, so its balanced class keeps moving [7, 8]. The results are exact statements about the settlement rule and about the carrier geometry.

Quadrupole carrier. Let b_i be the centered totals of the record counts at the two ports of antipodal pair i, so that $b \in W_5 = \{b \in \mathbb{R}^6 : \sum_i b_i = 0\}$, and set $T(b) = \sum_i b_i (n_i n_i^{\mathsf{T}} - I_3/3)$ for unit axis vectors n_i. The icosahedral inner products give $T^{*}T = (4/5)P_5$ with $P_5 = I_6 - \mathbf{1}\mathbf{1}^{\mathsf{T}}/6$, so W_5 is the geometric quadrupole carrier, irreducible under A_5. A supplied face axis has normalizer D_3, with $W_5|_{D_3} = \mathbf{1} \oplus \mathbf{2} \oplus \mathbf{2}$, whereas the physical geometric triplet restricts as sign $\oplus$ $\mathbf{2}$. The full native 30-seam port Laplacian acts on W_5 as $6I_5$, so a D_3 jump process on W_5 is a further dynamical hypothesis. With $\mathrm{Tr}(T_a T_b) = \delta_{ab}/2$, the color observable $A = (\sum_a T_a^2) \otimes P_0$ on $\mathbb{C}^3 \otimes W_5$, for the canonical rank-one face-axis projector P_0, has $\mathrm{Tr}\, A/15 = 4/15$ and $\mathrm{Tr}\, A^2/15 = 16/45$. Its identification as a mass response is a physical hypothesis.

Settled endpoints. The settlement rule transfers one record across a seam only if $N_i - N_j \geq 2$; it preserves $\sum_i N_i$ and strictly lowers $\sum_i N_i^2$, so every bounded fiber is finite and forward invariant. In the sector $N_i \in \{0, 1, 2\}$, $\sum_i N_i = 12$, repair schedules are orderings of maximal matchings of the active bipartite graph; exhaustive enumeration gives 73,789 states and 303 settled states in

nine rotational orbits, with largest minimum cost four and longest schedule six. Settledness implies $|N_i - N_j| \le d(i,j)$, and at totals divisible by twelve the 303 settled states exhaust all signed-integer endpoints. Their quadrupole spectra are $\{0,0,0\}$, $\{-2/\sqrt{5}, 0, 2/\sqrt{5}\}$, $\{-1-1/\sqrt{5}, 1-1/\sqrt{5}, 2/\sqrt{5}\}$ and its negative, so under an affine log-mass readout a nondegenerate ordered spectrum has adjacent log-gap ratio in $\{1, R, R^{-1}\}$, $R = (3/\sqrt{5}-1)/2$. At arbitrary total, the settled states reduce to 6,077 classes modulo uniform shift with exactly 31 quadrupole spectra, every integer settled state obeys $\operatorname{Tr} Q^2 \le 24/5$, and the shape invariant $\chi = (\operatorname{Tr} Q^3)^2/(\operatorname{Tr} Q^2)^3$ takes eleven exact values, giving nineteen admissible log-gap ratios. These restrictions select no physical total, endpoint, scale or attachment.

A driven record state. An additional renewal protocol appends one of the twenty-four signed port events uniformly after each complete settlement and settles again with fresh uniform proposals. On the 6,077 settled relative configurations it defines an irreducible kernel of period two with a unique positive stationary law, $\mathbb{E}_\pi Q = 0$, all odd centered moments zero, and $\mathbb{E}_\pi \operatorname{tr} Q^2 \simeq 1.4647638612$. Stationarity of the bounded cost $W(N) = \sum_i (N_i - S/12)^2$, which rises by $11/12$ per preparation on average and falls by two per accepted repair, gives exactly $11/24$ accepted repairs per preparation. The process is not reversible; it is a driven stationary record law, not a thermal equilibrium or quantum vacuum, and its moments transfer to a physical response only through an identification.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- A5 flavor potentials and residual-stabilizer selection [40, 59].
- Burnside_orbit_counting [42, 32].
- CAR_exterior_square_representation [31].
- Classical electroweak symmetry breaking and Goldstone counting in the Standard Model Higgs sector [95, 82, 49].
- Classical Koide relation, standard one-loop running and the existing OPH certified alpha and tau artifacts [60, 10, 82].
- complex_pole_to_running_width_Breit_Wigner_conversion [26, 88].
- cumulant expansions [61].
- De Broglie internal-clock and Zitterbewegung phase kinematics [38, 51, 86].
- Electroweak phenomenology, exterior weak multiplicities, hierarchy-problem literature, and commuting-square naturality diagnostics [16, 25, 90, 91, 67].
- exact_quadratic_field_linear_algebra [34].
- exact_rational_linear_algebra [92, 27].
- Exterior or Fock algebra and Standard Model exterior embeddings [25, 31, 45].
- exterior_algebra_and_CAR [31, 55].
- Finite clock models, Hamiltonian scaling, and determinant-line mass invariants [81, 52].
- Finite-group heat kernels [39, 85].

- finite_absorbing_Markov_chains_and_triangular_spectra [57, 52].
- finite_dimensional_min_max_and_rank_one_resolvent [28, 56].
- finite_group_invariant_operator_averaging [87].
- finite_group_representation_theory_and_Schur_lemma [87, 44].
- finite_irreducible_periodic_Markov_chain_stationarity [62, 77].
- finite_random_unitary_channels_and_class_generators [94, 39].
- Flavor-basis invariants, singular-value decompositions, and parameter non-identifiability [54, 29, 83].
- Free plane-wave evolution and de Broglie phase kinematics under translation flows [38, 96].
- Georgi–Jarlskog/Clebsch flavor textures, one-loop Yukawa running, and lattice light-quark ratios [46, 66, 14].
- Gibbs_relative_entropy_identity_and_finite_KMS [31].
- Harremoes and Topsoe (2001), DOI 10.1109/18.959272, Shannon entropy at fixed index of coincidence [12].
- Hellmann_Feynman_identity_and_spectral_perturbation [56].
- Koide's charged-lepton relation, Hermitian C3 circulants, and finite GNS constructions [60, 36, 30].
- Kraus_complete_positivity_and_trace_distance_contraction [94].
- maximal_matching_combinatorics [63].
- moment_and_cumulant_algebra [74, 80].
- numerical_quadrature_and_coupled_ODE_shooting [89, 37].
- Pauli algebra sigma_a sigma_b = delta_ab + i epsilon_abc sigma_c and the Pauli Hamiltonian derivation of g = 2 [79, 84].
- PDG and FLAG quark summaries [16, 14].
- perturbative renormalization-group transport [65, 64].
- positive_Hermitian_kinetic_normalization [52, 58].
- Primary mathematical and physical sources [25, 76].
- QCD running-to-pole conversion [33, 75].
- repulsively_bound_lattice_pairs [97].
- Residual-symmetry flavor mixing and icosahedral axis geometry [59, 40].
- scalar_ODE_parameter_sensitivity_and_comparison_barriers [50, 93].
- Schur_complement_differentiation [99, 68].
- Signed graph Laplacians, integer positive-definite matrices, and spectral residual estimates [98, 53, 78].
- SMDR_1_3_complex_pole_and_local_charged_current_matching [73, 72].
- Standard Model Higgs near-criticality calculations [43, 17].
- Standard_Model_beta_functions_top_only_specialization [66, 64].
- Standard_Model_two_loop_beta_functions_and_effective_potential [64, 41, 69].

- standard_SU3_fundamental_Casimir [47, 82].
- strict_convexity_and_exponential_family_KL_projection [35, 24].
- SU3_and_SU2_intertwiner_identities [47, 44].
- Tree-level charged-current couplings g/sqrt(2) with weak-isospin raising and lowering operators [95, 82].
- Tree-level electroweak neutral-current couplings Q = T3 + Y and T3 - Q sin^2 theta [48, 82, 95].
- TSIL_and_3VIL_loop_integral_evaluation [70, 71].

References

[1] B. Müller et al., *Observer-Patch Holography: formal proofs and executable receipts*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/Lean/Screen

[2] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, and D. A. Friedman, *Observers Are All You Need*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/observers_are_all_you_need.pdf

[3] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/reality_as_consensus_protocol.pdf

[4] B. Müller, A. Osika, K. Xue, B. Cassie, B. de La Fournière, and D. A. Friedman, *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/screen_microphysics_and_observer_synchronization.pdf

[5] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[6] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[7] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/executable_observer_construction.pdf

[8] FloatingPragma, *Repair laws: what is fixed, what is free and how to tell*, 2026. https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/model/CANDIDATES.md

[9] D. Tong, *Lectures on Quantum Field Theory*, Section 2, Free fields, University of Cambridge. https://www.damtp.cam.ac.uk/user/tong/qft/qfthtml/S2.html

[10] Y. Koide, *New view of quark and lepton mass hierarchy*, Physical Review D **28**, 252 (1983).

[11] C. A. Brannen, *The Lepton Masses*, preprint, 2006. http://brannenworks.com/MASSES2.pdf

[12] P. Harremoës and F. Topsøe, *Inequalities between entropy and index of coincidence derived from information diagrams*, IEEE Transactions on Information Theory **47**, 2944–2960 (2001). https://doi.org/10.1109/18.959272

[13] S. Antusch, K. Hinze, and S. Saad, *Updated Running Quark and Lepton Parameters at Various Scales*, arXiv:2510.01312, 2025. https://arxiv.org/abs/2510.01312

[14] Y. Aoki et al. (Flavour Lattice Averaging Group), *FLAG Review 2024*, arXiv:2411.04268, 2024, Tables 11–12. https://arxiv.org/abs/2411.04268

[15] S. Navas et al. (Particle Data Group), *CKM Quark-Mixing Matrix*, Review of Particle Physics 2024, Sec. 12.2.2, revised May 31, 2024. https://pdg.lbl.gov/2024/reviews/rpp2024-rev-ckm-matrix.pdf

[16] S. Navas et al. (Particle Data Group), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024), 2025 update, quark summary table. https://pdg.lbl.gov/2025/tables/rpp2025-sum-quarks.pdf

[17] D. Buttazzo et al., *Investigating the near-criticality of the Higgs boson*, JHEP **12** (2013) 089, Appendix B. https://arxiv.org/abs/1307.3536

[18] Particle Data Group, *Resonances*, Review of Particle Physics, 2025 update. https://pdg.lbl.gov/2025/reviews/rpp2025-rev-resonances.pdf

[19] D. Buchholz, *Gauss' law and the infraparticle problem*, Physics Letters B **174**, 331–334 (1986).

[20] P. Duch and W. Dybalski, *Infrared problem in quantum electrodynamics*, arXiv:2307.06114.

[21] M. Lüscher, *Abelian chiral gauge theories on the lattice with exact gauge invariance*, Nuclear Physics B **549**, 295–334 (1999), arXiv:hep-lat/9811032.

[22] M. Lüscher, *Lattice regularization of chiral gauge theories to all orders of perturbation theory*, Journal of High Energy Physics **06** (2000) 028, arXiv:hep-lat/0006014.

[23] D. Kadoh and Y. Kikukawa, *A simple construction of fermion measure term in $U(1)$ chiral lattice gauge theories with exact gauge invariance*, Journal of High Energy Physics **02** (2008) 063, arXiv:0709.3658.

[24] Amari, S. i., and Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society. https://doi.org/10.1090/mmono/191

[25] Baez, J., and Huerta, J. (2010). "The algebra of grand unified theories." *Bulletin of the American Mathematical Society* **47**, 483–552. https://doi.org/10.1090/S0273-0979-10-01294-2

[26] Bardin, D. Y., Leike, A., Riemann, T., and Sachwitz, M. (1988). "Energy-dependent width effects in e+e- annihilation near the Z-boson pole." *Physics Letters B* **206**, 539–542. https://doi.org/10.1016/0370-2693(88)91627-9

[27] Bareiss, E. H. (1968). "Sylvester's identity and multistep integer-preserving Gaussian elimination." *Mathematics of Computation* **22**, 565–578. https://doi.org/10.1090/S0025-5718-1968-0226829-0

[28] Bhatia, R. (1997). *Matrix Analysis*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0653-8

[29] Branco, G. C., Lavoura, L., and Silva, J. P. (1999). *CP Violation*. Clarendon Press, Oxford. ISBN 0-19-850399-7.

[30] Bratteli, O., and Robinson, D. W. (1987). *Operator Algebras and Quantum Statistical Mechanics 1: C*- and W*-Algebras, Symmetry Groups, Decomposition of States*. Springer. https://doi.org/10.1007/978-3-662-02520-8

[31] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2*. Springer. https://doi.org/10.1007/978-3-662-03444-6

[32] Burnside, W. (1911). *Theory of Groups of Finite Order*. Cambridge University Press. https://doi.org/10.1017/CBO9781139237253

[33] Chetyrkin, K., and Steinhauser, M. (2000). "The relation between the $\overline{\text{MS}}$ and the on-shell quark mass at order α_s^3." *Nuclear Physics B* **573**, 617–651. https://doi.org/10.1016/s0550-3213(99)00784-1

[34] Cohen, H. (1993). *A Course in Computational Algebraic Number Theory*. Springer. https://doi.org/10.1007/978-3-662-02945-9

[35] Csiszár, I. (1975). "I-divergence geometry of probability distributions and minimization problems." *Annals of Probability* **3**, 146–158. https://doi.org/10.1214/aop/1176996454

[36] Davis, P. J. (1979). *Circulant Matrices*. Wiley, New York. ISBN 0-471-05771-1.

[37] Davis, P. J., and Rabinowitz, P. (1984). *Methods of Numerical Integration*. Academic Press. https://doi.org/10.1016/C2013-0-10566-1

[38] de Broglie, L. (1925). "Recherches sur la théorie des quanta." *Annales de Physique* **10**, 22–128. https://doi.org/10.1051/anphys/192510030022

[39] Diaconis, P. (1988). *Group Representations in Probability and Statistics*. Institute of Mathematical Statistics (Lecture Notes–Monograph Series). https://doi.org/10.1214/lnms/1215467407

[40] Everett, L. L., and Stuart, A. J. (2009). "Icosahedral (A5) family symmetry and the golden ratio prediction for solar neutrino mixing." *Physical Review D* **79**, 085005. https://doi.org/10.1103/PhysRevD.79.085005

[41] Ford, C., Jack, I., and Jones, D. (1992). "The standard model effective potential at two loops." *Nuclear Physics B* **387**, 373–390. https://doi.org/10.1016/0550-3213(92)90165-8

[42] Frobenius, G. (1887). "Ueber die Congruenz nach einem aus zwei endlichen Gruppen gebildeten Doppelmodul." *Journal für die reine und angewandte Mathematik* **101**, 273–299. https://doi.org/10.1515/crll.1887.101.273

[43] Froggatt, C., and Nielsen, H. (1996). "Standard Model criticality prediction: top mass 173 ± 5 GeV and Higgs mass 135 ± 9 GeV." *Physics Letters B* **368**, 96–102. https://doi.org/10.1016/0370-2693(95)01480-2

[44] Fulton, W., and Harris, J. (1991). *Representation Theory: A First Course*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0979-9

[45] Georgi, H., and Glashow, S. L. (1974). "Unity of All Elementary-Particle Forces." *Physical Review Letters* **32**, 438–441. https://doi.org/10.1103/PhysRevLett.32.438

[46] Georgi, H., and Jarlskog, C. (1979). "A new lepton-quark mass relation in a unified theory." *Physics Letters B* **86**, 297–300. https://doi.org/10.1016/0370-2693(79)90842-6

[47] Georgi, H. (1999). *Lie Algebras in Particle Physics: From Isospin to Unified Theories*. Perseus Books. ISBN 0-7382-0233-9.

[48] Glashow, S. L. (1961). "Partial-symmetries of weak interactions." *Nuclear Physics* **22**, 579–588. https://doi.org/10.1016/0029-5582(61)90469-2

[49] Goldstone, J., Salam, A., and Weinberg, S. (1962). "Broken Symmetries." *Physical Review* **127**, 965–970. https://doi.org/10.1103/PhysRev.127.965

[50] Hartman, P. (2002). *Ordinary Differential Equations*. Society for Industrial and Applied Mathematics (Classics in Applied Mathematics). https://doi.org/10.1137/1.9780898719222

[51] Hestenes, D. (1990). "The zitterbewegung interpretation of quantum mechanics." *Foundations of Physics* **20**, 1213–1232. https://doi.org/10.1007/BF01889466

[52] Horn, R. A., and Johnson, C. R. (2012). *Matrix Analysis*. Cambridge University Press. https://doi.org/10.1017/CBO9781139020411

[53] Hou, Y., Li, J., and Pan, Y. (2003). "On the Laplacian Eigenvalues of Signed Graphs." *Linear and Multilinear Algebra* **51**, 21–30. https://doi.org/10.1080/0308108031000053611

[54] Jarlskog, C. (1985). "Commutator of the Quark Mass Matrices in the Standard Electroweak Model and a Measure of Maximal CP Nonconservation." *Physical Review Letters* **55**, 1039–1042. https://doi.org/10.1103/PhysRevLett.55.1039

[55] Jordan, P., and Wigner, E. (1928). "Über das Paulische Äquivalenzverbot." *Zeitschrift für Physik* **47**, 631–651. https://doi.org/10.1007/BF01331938

[56] Kato, T. (1995). *Perturbation Theory for Linear Operators*. Springer, Berlin (Classics in Mathematics). https://doi.org/10.1007/978-3-642-66282-9

[57] Kemeny, J. G., and Snell, J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics). ISBN 0-387-90192-2.

[58] King, S., and Peddie, I. (2004). "Canonical normalisation and Yukawa matrices." *Physics Letters B* **586**, 83–94. https://doi.org/10.1016/j.physletb.2004.01.087

[59] King, S. F., and Luhn, C. (2013). "Neutrino mass and mixing with discrete symmetry." *Reports on Progress in Physics* **76**, 056201. https://doi.org/10.1088/0034-4885/76/5/056201

[60] Koide, Y. (1983). "A fermion-boson composite model of quarks and leptons." *Physics Letters B* **120**, 161–165. https://doi.org/10.1016/0370-2693(83)90644-5

[61] Kubo, R. (1962). "Generalized cumulant expansion method." *Journal of the Physical Society of Japan* **17**, 1100–1120. https://doi.org/10.1143/JPSJ.17.1100

[62] Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*. American Mathematical Society. https://doi.org/10.1090/mbk/107

[63] Lovász, L., and Plummer, M. D. (2009). *Matching Theory*. AMS Chelsea Publishing, Providence. https://doi.org/10.1090/chel/367

[64] Luo, M., Wang, H., and Xiao, Y. (2003). "Two-loop renormalization group equations in general gauge field theories." *Physical Review D* **67**, 065019. https://doi.org/10.1103/PhysRevD.67.065019

[65] Machacek, M. E., and Vaughn, M. T. (1983). "Two-loop renormalization group equations in a general quantum field theory." *Nuclear Physics B* **222**, 83–103. https://doi.org/10.1016/0550-3213(83)90610-7

[66] Machacek, M. E., and Vaughn, M. T. (1984). "Two-loop renormalization group equations in a general quantum field theory (II). Yukawa couplings." *Nuclear Physics B* **236**, 221–232. https://doi.org/10.1016/0550-3213(84)90533-9

[67] Mac Lane, S. (1998). *Categories for the Working Mathematician*. Springer. https://doi.org/10.1007/978-1-4757-4721-8

[68] Magnus, J. R., and Neudecker, H. (2019). *Matrix Differential Calculus with Applications in Statistics and Econometrics*. Wiley. https://doi.org/10.1002/9781119541219

[69] Martin, S. P. (2002). "Two-loop effective potential for a general renormalizable theory and softly broken supersymmetry." *Physical Review D* **65**, 116003. https://doi.org/10.1103/PhysRevD.65.116003

[70] Martin, S. P., and Robertson, D. G. (2006). "TSIL: a program for the calculation of two-loop self-energy integrals." *Computer Physics Communications* **174**, 133–151. https://doi.org/10.1016/j.cpc.2005.08.005

[71] Martin, S. P., and Robertson, D. G. (2017). "Evaluation of the general three-loop vacuum Feynman integral." *Physical Review D* **95**, 016008. https://doi.org/10.1103/PhysRevD.95.016008

[72] Martin, S. P. (2019). "Matching relations for decoupling in the standard model at two loops and beyond." *Physical Review D* **99**, 033007. https://doi.org/10.1103/PhysRevD.99.033007

[73] Martin, S. P., and Robertson, D. G. (2019). "Standard Model parameters in the tadpole-free pure MS-bar scheme." *Physical Review D* **100**, 073004. https://doi.org/10.1103/PhysRevD.100.073004

[74] McCullagh, P. (1987). *Tensor Methods in Statistics*. Chapman and Hall, London. ISBN 0-412-27480-9.

[75] Melnikov, K., and van Ritbergen, T. (2000). "The three-loop relation between the MS-bar and the pole quark masses." *Physics Letters B* **482**, 99–108. https://doi.org/10.1016/S0370-2693(00)00507-4

[76] Nielsen, M. A., and Chuang, I. L. (2010). *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press. https://doi.org/10.1017/CBO9780511976667

[77] Norris, J. R. (1997). *Markov Chains*. Cambridge University Press. https://doi.org/10.1017/CBO9780511810633

[78] Parlett, B. N. (1998). *The Symmetric Eigenvalue Problem*. Society for Industrial and Applied Mathematics. https://doi.org/10.1137/1.9781611971163

[79] Pauli, W. (1927). "Zur Quantenmechanik des magnetischen Elektrons." *Zeitschrift für Physik* **43**, 601–623. https://doi.org/10.1007/BF01397326

[80] Peccati, G., and Taqqu, M. S. (2011). *Wiener Chaos: Moments, Cumulants and Diagrams*. Springer, Milan (Bocconi & Springer Series). https://doi.org/10.1007/978-88-470-1679-8

[81] Peres, A. (1980). "Measurement of time by quantum clocks." *American Journal of Physics* **48**, 552–557. https://doi.org/10.1119/1.12061

[82] Peskin, M. E., and Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley. https://doi.org/10.1201/9780429503559

[83] Rothenberg, T. J. (1971). "Identification in Parametric Models." *Econometrica* **39**, 577–591. https://doi.org/10.2307/1913267

[84] Sakurai, J. J., and Napolitano, J. (2020). *Modern Quantum Mechanics*. Cambridge University Press. https://doi.org/10.1017/9781108587280

[85] Saloff-Coste, L. (2004). "Random Walks on Finite Groups." In *Probability on Discrete Structures*, 263–346. Springer, Berlin (Encyclopaedia of Mathematical Sciences). https://doi.org/10.1007/978-3-662-09444-0_5

[86] Schrödinger, E. (1930). "Über die kräftefreie Bewegung in der relativistischen Quantenmechanik." *Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, 418–428. https://zbmath.org/?q=an:56.0754.06

[87] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[88] Sirlin, A. (1991). "Theoretical considerations concerning the Z0 mass." *Physical Review Letters* **67**, 2127–2130. https://doi.org/10.1103/PhysRevLett.67.2127

[89] Stoer, J., and Bulirsch, R. (2002). *Introduction to Numerical Analysis*. Springer, New York (Texts in Applied Mathematics). https://doi.org/10.1007/978-0-387-21738-3

[90] Susskind, L. (1979). "Dynamics of spontaneous symmetry breaking in the Weinberg-Salam theory." *Physical Review D* **20**, 2619–2625. https://doi.org/10.1103/PhysRevD.20.2619

[91] 't Hooft, G. (1980). "Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking." In *Recent Developments in Gauge Theories*, 135–157. Springer. https://doi.org/10.1007/978-1-4684-7571-5_9

[92] von zur Gathen, J., and Gerhard, J. (2013). *Modern Computer Algebra*. Cambridge University Press. https://doi.org/10.1017/CBO9781139856065

[93] Walter, W. (1970). *Differential and Integral Inequalities*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-86405-6

[94] Watrous, J. (2018). *The Theory of Quantum Information*. Cambridge University Press. https://doi.org/10.1017/9781316848142

[95] Weinberg, S. (1967). "A model of leptons." *Physical Review Letters* **19**, 1264–1266. https://doi.org/10.1103/PhysRevLett.19.1264

[96] Weinberg, S. (1995). *The Quantum Theory of Fields, Volume 1: Foundations*. Cambridge University Press. https://doi.org/10.1017/CBO9781139644167

[97] Winkler, K., Thalhammer, G., Lang, F., Grimm, R., Hecker Denschlag, J., Daley, A. J., et al. (2006). "Repulsively bound atom pairs in an optical lattice." *Nature* **441**, 853–856. https://doi.org/10.1038/nature04918

[98] Zaslavsky, T. (1982). "Signed graphs." *Discrete Applied Mathematics* **4**, 47–74. https://doi.org/10.1016/0166-218X(82)90033-6

[99] Zhang, F. (2005). *The Schur Complement and Its Applications*. Springer. https://doi.org/10.1007/b105056

[100] L. Susskind, "Lattice fermions," *Phys. Rev. D* **16**, 3031–3039 (1977).

[101] T. Misumi, T. Onogi and T. Yamaoka, "Taste-splitting mass and edge modes in 3+1D staggered fermions," arXiv:2604.02078v1 (2026), Sections 2.1–2.2.

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